Risk-Aware Action Repetition via Expected Skip Evaluation

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Abstract

Open-loop action repetition (also referred to as a skip) enables reinforcement learning agents to reduce decision frequency by repeating a selected action for an adaptively chosen duration. A key challenge is how to evaluate the skip value. Existing methods typically bootstrap from the greedy value of the skip's terminal state, implicitly assuming that the agent can immediately resume greedy control after the skip. This assumption is unreliable during training, when the underlying action policy is stochastic and inaccurate. We propose expected skip evaluation, which values a skip's terminal state through expected future control under a skip-evaluation policy rather than idealized greedy control. This reformulation accounts for the risk of imperfect future control during training while preserving the optimal skip-value fixed point as this policy approaches optimality. We incorporate this principle into *RARe (Risk-Aware Repetition)*, a practical framework for open-loop action repetition. Across six goal-reaching grid worlds and four continuous-control environments, *RARe* performs best or comparably to the best method in reward AUC. Furthermore, in grid worlds where action values and optimal decisions can be computed directly, *RARe* consistently incurs lower risk per step and more frequently makes risk-aware optimal decisions than open-loop baselines. Code is available at `https://github.com/funny-rl/RARe` and via a persistent DOI.

1 Introduction

Deep Reinforcement Learning (RL) has achieved remarkable success across a wide range of domains [19, 3]. However, standard Markov Decision Processes (MDPs) require agents to make decisions at every environment step, even when frequent control is unnecessary. This fixed decision frequency can hinder exploration and slow value propagation in long-horizon tasks [4]. Temporal abstraction addresses this limitation by allowing agents to commit to temporally extended courses of action rather than selecting a primitive action at every step [21, 22]. Among such approaches, open-loop action repetition methods learn adaptive skip lengths, executing the same primitive action for a selected duration to adjust decision frequency according to the current state and accelerate learning [11, 18, 4, 12].

A key question is how to estimate the value of a skip length. TempoRL [4] utilizes Skip-MDPs to formalize a skip as a temporally extended transition induced by repeated application of the same action for multiple steps. Consequently, a skip-value function should account for the rewards accumulated during execution and the action values at the skip's terminal state. However, TempoRL [4] and UTE [12] bootstrap from the maximum of these action values, implicitly assuming that the agent can immediately resume greedy action selection after the skip.

This assumption is often too strong because the skip policy that decides repetition lengths is learned while the action policy that selects primitive actions remains inaccurate and stochastic. Such imperfect

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40th Conference on Neural Information Processing Systems (NeurIPS 2026).

future control during training can cause an expected value loss after a skip and increase the probability of entering hazardous states. We define this value loss as risk: near hazards, it reflects the probability that stochastic action selection enters a failure state, drawing on the failure-based notion of risk [9, 8]. A skip should therefore be evaluated not only by the greedy action value at the terminal state, but also by how reliably the agent can control the system after the skip ends. Long skips can improve value propagation, but may become harmful when they end near hazards or in narrow passages where precise correction is required. Thus, skip evaluation should distinguish terminal states with high greedy action values from those that also retain high expected value under the skip-evaluation policy, the policy used to evaluate the state where a skip ends.

In this work, we propose *RARe (Risk-Aware Repetition)*, an open-loop action repetition framework based on expected skip evaluation. By evaluating skip-terminal states under the skip-evaluation policy rather than greedy control, RARe reflects imperfect next-action selection during learning. As this policy becomes greedy, the objective recovers the standard greedy skip-value objective. RARe promotes useful temporal commitments while discouraging skips whose terminal states require unreliable future control. We summarize our contributions as follows:

- We identify the risk neglected by greedy terminal bootstrapping when future action selection is imperfect during training.
- We address the limitation of greedy terminal bootstrapping during training through expected skip evaluation and incorporate this approach into RARe, while recovering the greedy skip-value objective as the skip-evaluation policy becomes greedy.
- We evaluate RARe on six goal-reaching grid worlds and four continuous-control environments, where it performs best or comparably to the best method in reward AUC and, in grid worlds, incurs lower risk per step and makes risk-aware optimal decisions more frequently than open-loop baselines.

2 Background

In this section, we review MDPs and Skip-MDPs, and then define the risk of imperfect control after a skip.

2.1 Markov Decision Processes

RL problems are typically formalized as MDPs [2], defined by a tuple $\mathcal{M} := \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where $\mathcal{S}$ is the state space and $\mathcal{A}$ is the action space. The transition dynamics are given by $\mathcal{P}^a_{ss'} = P(s_{t+1} = s' \mid s_t = s, a_t = a)$, which represents the probability of moving to state s' after taking action a in state s. The reward function $\mathcal{R}^a_s = \mathbb{E}[r_t \mid s_t = s, a_t = a]$ denotes the expected immediate reward, and $\gamma \in (0, 1)$ is the discount factor. A policy $\pi(a \mid s)$ induces the expected return $\mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t\right]$, and the goal of the agent is to find an optimal policy that maximizes this return.

The action-value function of π is defined as:

$$Q^\pi(s, a) = \mathbb{E}\left[r_t + \gamma Q^\pi(s_{t+1}, a_{t+1}) \mid s_t = s, a_t = a\right], \tag{1}$$

and the optimal action-value function is defined as $Q^*(s, a) = \max_\pi Q^\pi(s, a)$. The corresponding optimal state-value function is denoted as $V^*(s) := \max_{a \in \mathcal{A}} Q^*(s, a)$.

2.2 Skip-MDPs

Skip-MDPs [4] formalize action repetition as skip-connections between states. A Skip-MDP $\mathcal{M}_\mathcal{C}$ contextualizes an original MDP $\mathcal{M}$ with a set $\mathcal{C}$ of skip-connections $c = \langle s, a, j \rangle$, where $j \in \mathcal{J} = \{1, \ldots, J\}$ is the skip length. A skip-connection exists if a state s_{t+j} is reachable from s_t by repeating the same action a_t for j consecutive time steps. Each skip-connection induces a skip transition function $\mathcal{P}_c$ and a skip reward function $\mathcal{R}_c$:

$$\mathcal{P}_c(s_t, a_t, s_{t+j}) = \sum_{s_{t+1} \ldots s_{t+j-1}} \prod_{k=0}^{j-1} \mathcal{P}^{a_t}_{s_{t+k} s_{t+k+1}}, \quad \mathcal{R}_c(s_t, a_t, j) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} \,\middle|\, s_t, a_t, j\right].$$

For $j = 1$, these functions recover the original $\mathcal{P}^a_{ss'}$ and $\mathcal{R}^a_s$. A skip policy $\pi_J(j \mid s, a)$ selects a skip length for each state-action pair, which can accelerate value propagation and reduce decision frequency. The skip-value function gives the expected return of repeating a for j steps and then following π:

$$\tilde{Q}^{\pi_J}(s, j \mid a) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j Q^\pi(s_{t+j}, a_{t+j}) \;\middle|\; s_t = s, a, j\right], \tag{2}$$

and the optimal skip-value function is defined as

$$\tilde{Q}^*(s, j \mid a) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j V^*(s_{t+j}) \;\middle|\; s_t = s, a, j\right].$$

Using this formulation, TempoRL [4] replaces the bootstrap value at s_{t+j} with the greedy value of the learned action-value function Q, giving the skip-value function:

$$\tilde{Q}^{\pi_J}_{\text{TempoRL}}(s, j \mid a) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j \max_{a' \in \mathcal{A}} Q(s_{t+j}, a') \;\middle|\; s_t = s, a, j\right]. \tag{3}$$

2.3 Formalism of Risk in Skip-MDPs

To motivate the risk-aware perspective of this study, we formalize the risk arising from imperfect future control under stochastic action selection. We quantify this risk as the expected value loss of control under a policy π relative to idealized greedy control at the resulting state s_{t+j}:

$$\textit{Risk}(s_t, a_t, j) := \mathbb{E}_{s_{t+j} \sim \mathcal{P}_c(s_t, a_t, \cdot)}\left[\max_{a' \in \mathcal{A}} Q(s_{t+j}, a') - \mathbb{E}_{a' \sim \pi(\cdot \mid s_{t+j})}[Q(s_{t+j}, a')]\right]. \tag{4}$$

The right-hand side of Eq. (4) consists of two terms. The first term, $\max_{a' \in \mathcal{A}} Q(s_{t+j}, a')$, is the greedy value assumed by Eq. (3), which presumes that the agent resumes greedy control at s_{t+j}. The second term, $\mathbb{E}_{a' \sim \pi(\cdot \mid s_{t+j})}[Q(s_{t+j}, a')]$, is the expected value of the action actually selected by the policy π after the skip. Since the maximum of the action values is never smaller than their expectation under π, the risk is theoretically guaranteed to be non-negative: $\textit{Risk}(s_t, a_t, j) \geq 0$.

3 Theory of Skip Evaluation

In order to motivate the transition to expected skip evaluation, we analyze the theoretical and training-time problems inherent in the conventional greedy skip evaluation approach.

3.1 Risk in the Training Phase

During training, the agent acts under a stochastic policy π, where optimal actions are not always selected. Assume that at each decision point, there is a non-zero probability $p > 0$ of selecting a suboptimal action that leads to a hazardous state. Then, over n consecutive decision points, the probability of encountering at least one hazardous state is $1 - (1 - p)^n$. Therefore, policies with frequent decision-making are more exposed to hazardous transitions, whereas longer skips can reduce this exposure by decreasing the number of decision points. The skip value in Eq. (2) reflects this exposure because it follows π after the skip, whereas the TempoRL value in Eq. (3) assumes greedy control at s_{t+j} and does not.

Proposition 1 (Risk in a hazard-adjacent chain)**.** *Consider a Skip-MDP containing a hazard-adjacent chain, whose hazardous states incur a negative reward when entered. Let π and π_J denote the action and skip policies, both ϵ-greedy with $\epsilon > 0$. Then, there exist a state s, an action a, and skip lengths $j_1, j_2 \in \mathcal{J}$ with $j_2 > j_1$ such that*

$$\tilde{Q}^{\pi_J}(s, j_2 \mid a) > \tilde{Q}^{\pi_J}(s, j_1 \mid a).$$

Proposition 1 explains that, when an agent must repeatedly select the same action, stochastic re-decisions can increase hazard exposure. Thus, shorter skips can be penalized when they require

frequent re-decisions near hazardous states, while longer skips may be preferred by reducing exposure to such risky decisions. Importantly, this does not imply that longer skips are always optimal; it only establishes that they can be necessary when frequent decision-making induces risk. Therefore, risk-aware skip evaluation should adaptively balance short skips for precise control and long skips for reduced exposure to hazards. The full proof is provided in Appendix A.1.

3.2 Properties of Optimal Skip Values

In the deterministic setting considered below, once Q converges to Q^*, Eq. (3) reduces to $\tilde{Q}^*$, and longer skips provide no additional value over one-step evaluation.

Proposition 2. *Consider a deterministic MDP with a discount factor $\gamma \in (0, 1)$. Let $\tilde{Q}^*$ be the optimal skip-value function, defined as:*

$$\tilde{Q}^*(s, j \mid a) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j V^*(s_{t+j}) \;\middle|\; s_t = s, a, j\right], \tag{5}$$

where $V^*(s) := \max_{a \in \mathcal{A}} Q^*(s, a)$. Then, for all $j \geq 1$ ($j \in \mathcal{J}$), the following inequality holds:

$$\tilde{Q}^*(s, 1 \mid a) \geq \tilde{Q}^*(s, j \mid a). \tag{6}$$

Proposition 2 implies that, once the skip-value function reaches the optimal value, longer skips provide no additional value beyond $\tilde{Q}^*(s, 1 \mid a)$. If a skip length j' is optimal, then all shorter skips along the trajectory induced by repeating a attain the same value, i.e., $\tilde{Q}^*(s, j'' \mid a) = \tilde{Q}^*(s, 1 \mid a)$ for $1 \leq j'' \leq j'$. Thus, at optimality, multiple skip lengths become value-equivalent. Without an additional criterion, a neural approximation may break this tie arbitrarily due to approximation error or training noise, making the learned skip policy unstable or inefficient. Therefore, beyond preserving optimal values, a robust skip evaluation framework benefits from a training-time criterion that resolves this value-equivalence, establishing a risk-aware preference ordering among otherwise equivalent skip lengths. The full proof is provided in Appendix A.2.

3.3 Expected Skip Evaluation

To address the limitations of greedy skip evaluation and resolve the training-time tie-breaking problem, we propose a principled alternative that replaces the greedy bootstrap operator with an expectation over future actions. Specifically, we define the skip-value function as:

$$\tilde{Q}^{\pi_J}_{\text{RARe}}(s, j \mid a) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j \mathbb{E}_{a' \sim \pi_e(\cdot \mid s_{t+j})}\left[Q^{\pi_e}(s_{t+j}, a')\right] \;\middle|\; s_t = s, a, j\right]. \tag{7}$$

We define π_e as the skip-evaluation policy used to compute the TD target of the skip-value function. Importantly, π_e is decoupled from the behavior policy that generates trajectories: it may coincide with it or be chosen independently. This flexibility makes the formulation compatible with the standard off-policy learning setting and replay-buffer-based implementations [20].

Crucially, while greedy skip evaluation methods have no risk-based preference among value-equivalent skip lengths because they assume perfect future control, our expected skip evaluation dynamically breaks this value-equivalence during training. By evaluating the bootstrap target through the expectation $\mathbb{E}_{a' \sim \pi_e(\cdot \mid s_{t+j})}$, the bootstrap term equals the greedy bootstrap reduced by ***Risk***, where ***Risk*** follows Eq. (4) with $\pi = \pi_e$ and $Q = Q^{\pi_e}$. This reflects imperfect future control in the resulting state s_{t+j}. Consequently, during training, a long skip is not penalized merely for being long; rather, it receives a lower value when its terminal state requires unreliable or risky future control under π_e. This allows the skip policy to avoid stopping at such states: it can shorten a skip that would end near a hazard, or extend one to pass hazard-adjacent states without a re-decision (Proposition 1).

3.4 Convergence of Expected Skip Evaluation

We show that expected skip evaluation preserves the optimal skip-value function. Although replacing the $\max$ operator with an expectation alters the intermediate learning dynamics to inject

Algorithm 1 RARe: Risk-Aware Repetition

Require: behavior policy π, skip policy π_J, skip-evaluation policy π_e, (action, skip) buffers $\mathcal{D}, \tilde{\mathcal{D}}$

Initialize the base networks with their target networks, and skip critic $\tilde{Q}^{\pi_J}_{\text{RARe}}$
Set $Q^{\text{eval}} \leftarrow Q$ for RARe-sh, or initialize a separate network $Q^{\text{eval}} \leftarrow Q^{\pi_e}$ for RARe-se
for each training step **do**
    Collect experience with π and π_J, repeating each action for its selected skip length, and store it in $\mathcal{D}$ and $\tilde{\mathcal{D}}$
    Update Q on $\mathcal{D}$ with the base algorithm's TD target
    if RARe-se **then** update Q^{π_e} on $\mathcal{D}$ with target $r + \gamma \mathbb{E}_{a' \sim \pi_e(\cdot|s')}[Q^{\pi_e}_{\text{target}}(s', a')]$ **end if**
    Update $\tilde{Q}^{\pi_J}_{\text{RARe}}$ on $\tilde{\mathcal{D}}$ with target $\sum_{k=0}^{j-1} \gamma^k r_k + \gamma^j \mathbb{E}_{a_j \sim \pi_e(\cdot|s_j)}[Q^{\text{eval}}_{\text{target}}(s_j, a_j)]$
    Soft-update all target networks
end for

risk-awareness, the two targets become mathematically equivalent once the evaluation policy becomes greedy and optimal. We emphasize that this convergence requires π_e to converge to a deterministic optimal policy, independently of whether it matches the behavior policy; the risk-aware gap is a feature of training and vanishes only at optimality.

Proposition 3. *Let π_e be the evaluation policy used in expected skip evaluation, with action-value function Q^{π_e}. Suppose $Q^{\pi_e} \to Q^*$ and that π_e converges to a deterministic optimal policy π^* such that*

$$\pi_e(a^* \mid s) \to 1 \quad \textit{for every state } s \textit{ and some } a^* \in \arg\max_{a \in \mathcal{A}} Q^*(s, a).$$

Then, for any state s, action a, and skip length $j \in \mathcal{J}$,

$$\mathbb{E}_{a' \sim \pi_e(\cdot|s_{t+j})} \left[Q^{\pi_e}(s_{t+j}, a') \right] \to \max_{a'} Q^*(s_{t+j}, a'). \tag{8}$$

Consequently, under TD convergence assumptions, expected skip evaluation asymptotically converges to $\tilde{Q}^*$.

Proposition 3 shows that the expectation-based target modifies training-time skip evaluation while asymptotically preserving the optimal fixed point. As $Q^{\pi_e} \to Q^*$ and $\pi_e \to \pi^*$, $\mathit{Risk}(s_t, a_t, j) \to 0$ and the target recovers the standard optimal skip-value objective. The proof is given in Appendix A.3.

4 RARe: Risk-Aware Repetition

RARe keeps the base action-value update and changes only how skips are evaluated, learning skip values under imperfect future control rather than idealized greedy control. RARe has two variants: RARe-se learns a separate evaluation critic Q^{π_e} with its own target network, whereas RARe-sh shares the base critic Q to construct skip-value TD targets. Algorithm 1 summarizes the training pipeline.

4.1 Learning Framework

At each training step, RARe samples transitions $(s, a, r, s') \sim \mathcal{D}$ and skip transitions $(s, a, j, \{r_k\}_{k=0}^{j-1}, s_j) \sim \tilde{\mathcal{D}}$. The base critic Q is updated with the objective of the base algorithm. In discrete action spaces, we employ a DDQN [24] target:

$$y = r + \gamma Q_{\text{target}} \left(s', \arg\max_{a' \in \mathcal{A}} Q(s', a') \right), \tag{9}$$

and the behavior policy π is ϵ-greedy with respect to Q. In continuous action spaces, we use the DDPG [16] target, and π is the actor with exploration noise. RARe-se additionally optimizes the evaluation critic Q^{π_e} against $y^{\pi_e} = r + \gamma \mathbb{E}_{a' \sim \pi_e(\cdot|s')}[Q^{\pi_e}_{\text{target}}(s', a')]$; RARe-sh has no such update.

4.2 RARe Skip-Value Update

For each skip transition, RARe optimizes the skip-value function $\tilde{Q}^{\pi_J}_{\text{RARe}}$ using the expected target defined under the skip-evaluation policy π_e:

$$\tilde{y}^{\pi_J} = \sum_{k=0}^{j-1} \gamma^k r_k + \gamma^j \mathbb{E}_{a_j \sim \pi_e(\cdot|s_j)} \left[Q^{\text{eval}}_{\text{target}}(s_j, a_j) \right], \tag{10}$$

where $Q^{\text{eval}} = Q^{\pi_e}$ for RARe-se and $Q^{\text{eval}} = Q$ for RARe-sh. RARe-sh thus replaces Q^{π_e} in Eq. (7) with the base critic, which targets Q^* rather than Q^{π_e}. Both variants share the same optimal fixed point: Proposition 3 applies directly to RARe-se, and for RARe-sh, $Q \to Q^*$ and $\pi_e \to \pi^*$ jointly drive the discounted bootstrap term to $\gamma^j \max_{a'} Q^*(s_j, a')$.

The variants instead differ in how far the risk signal extends during training. In RARe-se, Q^{π_e} propagates imperfect control over the entire horizon beyond s_j. In RARe-sh, risk enters only through the one-step expectation at s_j, since Q assumes greedy control afterward.

4.3 Practical Estimation of the Expected Target

In discrete domains, π_e is an ϵ-greedy distribution induced by Q^{eval}, assigning $\epsilon/|\mathcal{A}|$ to all actions and the remaining $1 - \epsilon$ mass to the greedy choice, so the expectation is computed exactly:

$$\mathbb{E}_{a' \sim \pi_e(\cdot|s)} \left[Q^{\text{eval}}_{\text{target}}(s, a') \right] = \sum_{a' \in \mathcal{A}} \pi_e(a' \mid s) Q^{\text{eval}}_{\text{target}}(s, a'). \tag{11}$$

In continuous domains, π_e is a clipped Gaussian perturbation of the target actor, $\pi_{\text{target}}(s) + \xi$ with $\xi \sim \mathcal{N}(0, \sigma_e^2 I)$. Because $\mathcal{A}$ is uncountable, we approximate the expectation with N Monte Carlo samples:

$$\mathbb{E}_{a' \sim \pi_e(\cdot|s)} \left[Q^{\text{eval}}_{\text{target}}(s, a') \right] \approx \frac{1}{N} \sum_{i=1}^{N} Q^{\text{eval}}_{\text{target}} \left(s, \text{clip}(\pi_{\text{target}}(s) + \xi_i, -a_{\max}, a_{\max}) \right). \tag{12}$$

RARe-sh sets ϵ or σ_e to the behavior value, so π_e matches the behavior exploration level, whereas RARe-se uses an independent schedule. The two policies serve different roles: the behavior policy explores, while π_e specifies how reliable future control is assumed to be in the skip target. An independent schedule sets this assumption without changing exploration and can anneal π_e toward greedy, as Proposition 3 requires, on its own timescale.

5 Experiments

We evaluate RARe across six goal-reaching grid worlds, two standard continuous-control tasks, and two safety-critical continuous-control environments [10]. We compare against fixed-frequency baselines (DDQN [24] or DDPG [16]), open-loop repetition methods (TempoRL [4] and UTE [12]), and a closed-loop temporal abstraction baseline (TAAC [27]) when applicable. Environment and baseline details are given in Appendices B and C. All main results are computed over 20 random seeds and reported using the interquartile mean (IQM) [1] with 95% bootstrap confidence intervals. In all performance tables, bold font indicates the best-performing method for each environment, while gray shading marks our proposed method.

5.1 Evaluation Metrics

We measure task performance in all environments. On grid environments, where the dynamics are known, we also measure risk and decision quality directly. Full definitions are provided in Appendix D.

Normalized AUC Score. To reflect both learning speed and final performance, we integrate the normalized score over post-warm-up evaluation checkpoints $t_1, \ldots, t_K$:

$$\text{AUC}_{\text{norm}} = \frac{1}{t_K - t_1} \int_{t_1}^{t_K} \text{score}_{\text{norm}}(t)\, dt,$$

where $\text{score}_{\text{norm}}(t)$ rescales the evaluation reward between fixed reference scores.

Table 1: Normalized reward AUC on grid environments (higher is better).

Loop	Model	Standard			Randomized		
		Cliff	Bridge	ZigZag	RCliff	RBridge	RZigZag
✗	DDQN	$.522_{[.43,.61]}$	$.064_{[.00,.16]}$	$.499_{[.42,.55]}$	$.452_{[.32,.56]}$	$.095_{[.08,.11]}$	$.169_{[.14,.22]}$
Open	TempoRL	$.699_{[.66,.73]}$	$.753_{[.70,.79]}$	$.636_{[.59,.68]}$	$.730_{[.67,.78]}$	$.397_{[.36,.43]}$	$.295_{[.26,.33]}$
	UTE	$.696_{[.65,.72]}$	$.645_{[.58,.71]}$	$.567_{[.49,.63]}$	$.739_{[.68,.79]}$	$.322_{[.30,.35]}$	$.261_{[.20,.31]}$
	RARe-sh	$.685_{[.64,.72]}$	$\mathbf{.766}_{[.72,.80]}$	$.630_{[.56,.71]}$	$\mathbf{.763}_{[.72,.79]}$	$.474_{[.43,.51]}$	$.318_{[.28,.35]}$
	RARe-se	$\mathbf{.705}_{[.68,.73]}$	$.760_{[.70,.80]}$	$\mathbf{.685}_{[.62,.74]}$	$.740_{[.70,.77]}$	$\mathbf{.495}_{[.46,.52]}$	$\mathbf{.369}_{[.32,.41]}$

Table 2: Risk per step on grid environments, averaged over training (lower is better).

Model	Standard			Randomized		
	Cliff	Bridge	ZigZag	RCliff	RBridge	RZigZag
TempoRL	$3.70_{[3.5,3.9]}$	$2.73_{[2.6,3.0]}$	$2.61_{[2.3,3.1]}$	$4.82_{[4.5,5.0]}$	$7.64_{[7.4,8.0]}$	$5.61_{[5.4,5.9]}$
UTE	$4.35_{[4.1,4.7]}$	$3.04_{[2.8,3.2]}$	$3.36_{[3.2,4.0]}$	$5.31_{[5.0,5.7]}$	$7.75_{[7.4,8.2]}$	$6.15_{[5.9,6.5]}$
RARe-sh	$\mathbf{2.80}_{[2.6,3.0]}$	$1.94_{[1.8,2.1]}$	$\mathbf{1.95}_{[1.7,2.3]}$	$\mathbf{3.22}_{[3.0,3.6]}$	$\mathbf{6.34}_{[6.0,6.7]}$	$5.30_{[5.0,5.5]}$
RARe-se	$2.97_{[2.8,3.2]}$	$\mathbf{1.73}_{[1.6,1.9]}$	$2.14_{[1.9,2.5]}$	$3.99_{[3.6,4.5]}$	$6.50_{[6.1,7.0]}$	$\mathbf{5.12}_{[4.9,5.4]}$

Risk per Step. Let each evaluation trajectory $\tau = \{(s_n, a_n, j_n)\}_{n=1}^{N_\tau}$ of the deterministic evaluation policy take T_τ primitive steps. We measure

$$\mathrm{Risk}_{\mathrm{step}} = \underbrace{\frac{\sum_\tau N_\tau}{\sum_\tau T_\tau}}_{\text{decisions per step}} \cdot \underbrace{\frac{\sum_\tau \sum_{n=1}^{N_\tau} \mathit{Risk}(s_n, a_n, j_n)}{\sum_\tau N_\tau}}_{\text{Risk per decision}} = \frac{\sum_\tau \sum_{n=1}^{N_\tau} \mathit{Risk}(s_n, a_n, j_n)}{\sum_\tau T_\tau},$$

where *Risk* is Eq. (4) with the exact Q^π of the behavior ϵ-greedy policy frozen at the checkpoint, for every method, and is zero at terminal endpoints. The decomposition shows that it grows with both how often control returns to the policy and how much value is lost at each endpoint. We divide by primitive steps because a per-episode sum would fall whenever an episode ends early in a hazard.

Risk-aware Optimal Decision Ratio. The Risk-aware Optimal Decision Ratio (RODR) measures whether decisions follow risk-aware optimal behavior:

$$\mathrm{RODR}(\tau) = \frac{1}{N_\tau} \sum_{n=1}^{N_\tau} \mathbb{I}\left(a_n \in \mathcal{A}^*(s_n) \ \wedge \ j_n \in \mathcal{D}^*(s_n, a_n) \ \wedge \ d(e_n) < \min_{s \in \mathcal{V}_n} d(s) \right),$$

where d is the shortest safe distance to the goal, $\mathcal{A}^*$ contains the actions that reduce d, $\mathcal{D}^*$ contains the skip lengths that make maximal safe progress, which reduces decision points as motivated by Proposition 1, e_n is the endpoint of decision n, and $\mathcal{V}_n$ is the set of states visited before e_n. An oracle-matching decision is credited only if its endpoint is closer to the goal than every previously visited state. Unlike Risk, RODR does not depend on Q or ϵ, so low Risk with high RODR indicates that risk is reduced without sacrificing progress. Both metrics are computed at each checkpoint, with RODR averaged over evaluation episodes, and summarized with the same integral as $\mathrm{AUC}_{\mathrm{norm}}$.

5.2 Grid-world Environments

We first evaluate RARe on six grid-world environments with hazardous or restricted optimal paths: *CliffWalking*, *Bridge*, *ZigZag*, and their randomized versions, which resample the start and goal at every reset. As shown in Table 1, a RARe variant achieves the highest normalized AUC in all six environments. On the standard layouts, both variants outperform DDQN and are never significantly worse than TempoRL or UTE. The advantage grows on the randomized tasks, where both variants significantly outperform TempoRL on *RBridge*, RARe-sh also on *RCliffWalking*, and RARe-se also on *RZigZag*. Across layouts, RARe separates most clearly on the narrow *Bridge* layouts, where most comparisons with TempoRL and UTE are significant, followed by *ZigZag*, where the gains are significant for RARe-se, whereas on *CliffWalking* the open-loop methods remain largely comparable.

Table 3: RODR on grid environments, averaged over training (higher is better).

Loop	Model	Standard			Randomized		
		Cliff	Bridge	ZigZag	RCliff	RBridge	RZigZag
✗	DDQN	$.122_{[.10, .14]}$	$.028_{[.02, .05]}$	$.157_{[.14, .17]}$	$.124_{[.09, .15]}$	$.031_{[.028, .034]}$	$.062_{[.06, .07]}$
Open	TempoRL	$.291_{[.27, .31]}$	$.287_{[.27, .30]}$	$.328_{[.29, .36]}$	$.329_{[.30, .35]}$	$.149_{[.14, .16]}$	$.135_{[.12, .15]}$
	UTE	$.236_{[.22, .25]}$	$.219_{[.20, .24]}$	$.248_{[.21, .28]}$	$.285_{[.26, .31]}$	$.127_{[.12, .14]}$	$.103_{[.08, .12]}$
	RARe-sh	$.375_{[.35, .40]}$	$.339_{[.31, .37]}$	$.398_{[.35, .44]}$	$\mathbf{.460}_{[.42, .48]}$	$.229_{[.21, .24]}$	$\mathbf{.175}_{[.16, .19]}$
	RARe-se	$\mathbf{.411}_{[.39, .44]}$	$\mathbf{.424}_{[.40, .44]}$	$\mathbf{.440}_{[.40, .48]}$	$.443_{[.41, .47]}$	$\mathbf{.357}_{[.33, .38]}$	$\mathbf{.175}_{[.16, .19]}$

Table 4: Normalized reward AUC on continuous environments (higher is better).

Loop	Model	Continuous Control		Safe RL	
		MountainCar	PointMaze	Goal	Button
✗	DDPG	$.017_{[.017, .017]}$	$.511_{[.382, .602]}$	$.298_{[.190, .456]}$	$.386_{[.312, .492]}$
Closed	TAAC	$.523_{[.258, .673]}$	$.604_{[.538, .660]}$	$.401_{[.249, .545]}$	$.327_{[.260, .400]}$
Open	TempoRL	$.674_{[.638, .696]}$	$.613_{[.567, .681]}$	$.588_{[.427, .742]}$	$.545_{[.471, .611]}$
	UTE	$.632_{[.564, .669]}$	$.607_{[.533, .658]}$	$.567_{[.431, .733]}$	$.636_{[.576, .705]}$
	RARe-sh	$.664_{[.626, .694]}$	$.642_{[.557, .697]}$	$.596_{[.481, .740]}$	$.597_{[.529, .681]}$
	RARe-se	$\mathbf{.675}_{[.649, .703]}$	$\mathbf{.687}_{[.566, .750]}$	$\mathbf{.685}_{[.523, .793]}$	$\mathbf{.656}_{[.540, .754]}$

Tables 2 and 3 reveal differences that reward alone does not capture. Both RARe variants incur lower Risk per step and achieve higher RODR than TempoRL and UTE in all six environments, with the paired 95% confidence interval excluding zero in every comparison. By the decomposition of Risk per step, this reduction can stem from fewer decisions per step, as motivated by Proposition 1, from lower Risk per decision, or from both. These gains accompany reward gains: on *RBridge*, RARe-se raises the normalized AUC of TempoRL from .397 to .495, reduces Risk per step from 7.64 to 6.50, and more than doubles RODR from .149 to .357. They also persist where rewards are close: on *CliffWalking*, RARe-se matches the reward of TempoRL while reducing Risk per step from 3.70 to 2.97 and raising RODR from .291 to .411. Thus, matching return does not imply matching decisions, and greedy skip evaluation leaves the agent more exposed under imperfect future control. Environment details and learning curves are provided in Appendices B.1 and E.1.

5.3 Continuous Environments

We next evaluate RARe in continuous action spaces, covering both exploration-oriented control tasks and safety-critical tasks with scalarized costs. As shown in Table 4, RARe-se achieves the highest normalized AUC in all four environments. On *MountainCar*, DDPG remains near the random baseline, whereas methods with temporal abstraction make progress under the same budget, indicating the value of multi-step commitment for early exploration; RARe-se and TempoRL are effectively tied on this task. On *PointMaze*, both RARe variants significantly exceed DDPG, whereas the paired 95% confidence intervals of TempoRL, UTE, and TAAC against DDPG include zero.

We also evaluate RARe on safety-critical continuous-control tasks from Safety-Gymnasium [10]. We use these tasks as stress tests for risk-aware temporal abstraction, rather than as constrained RL benchmarks, and compare all methods under the same scalarized reward:

$$\tilde{R}_t = w_r \cdot R_t - w_c \cdot C_t, \tag{13}$$

where R_t is the performance reward, C_t is the constraint cost, and $w_r, w_c > 0$ are scaling factors. This setting is challenging for action repetition because long skips can accelerate exploration in benign regions but amplify failures near hazards. Both RARe variants significantly exceed DDPG and TAAC on *Goal* and *Button*, whereas TempoRL and UTE significantly exceed TAAC only on *Button*. Across all four environments, both RARe variants separate from DDPG and TAAC more consistently than TempoRL and UTE do, and neither is significantly worse than any baseline. RARe-se also scores above RARe-sh on all four tasks, although not significantly. Environment details are provided in Appendices B.3 and B.4, and learning curves in Appendix E.2.

Table 5: TD-backup ablation of RARe on grid environments: normalized AUC (higher is better). Action and Skip give the backup of the action critic and of the skip value: G (greedy) or E (expected under the ϵ-greedy policy). Shared bootstraps skips from the action critic, so its G/G arm is TempoRL and its G/E arm RARe-sh; Separate uses a separate endpoint critic, and its G/E arm is RARe-se.

Critic	Act.	Skip	Standard			Randomized		
			Cliff	Bridge	ZigZag	RCliff	RBridge	RZigZag
Shared	E	G	$.442_{[.42, .47]}$	$.533_{[.52, .54]}$	$.436_{[.40, .46]}$	$.478_{[.39, .54]}$	$.143_{[.13, .16]}$	$.179_{[.14, .23]}$
	E	E	$.449_{[.42, .48]}$	$.525_{[.51, .54]}$	$.441_{[.41, .46]}$	$.482_{[.38, .56]}$	$.145_{[.13, .16]}$	$.142_{[.12, .17]}$
	TempoRL		$\mathbf{.699}_{[.66, .73]}$	$.753_{[.70, .79]}$	$\mathbf{.636}_{[.59, .68]}$	$.730_{[.67, .78]}$	$.397_{[.36, .43]}$	$.295_{[.26, .33]}$
	RARe-sh		$.685_{[.64, .72]}$	$\mathbf{.766}_{[.72, .80]}$	$.630_{[.56, .71]}$	$\mathbf{.763}_{[.72, .79]}$	$\mathbf{.474}_{[.43, .51]}$	$\mathbf{.318}_{[.28, .35]}$
Separate	E	G	$.465_{[.43, .49]}$	$.540_{[.52, .55]}$	$.454_{[.43, .47]}$	$.442_{[.38, .53]}$	$.123_{[.11, .14]}$	$.133_{[.10, .16]}$
	E	E	$.479_{[.45, .51]}$	$.541_{[.52, .55]}$	$.465_{[.44, .49]}$	$.467_{[.39, .54]}$	$.162_{[.14, .18]}$	$.152_{[.13, .18]}$
	G	G	$.688_{[.65, .73]}$	$.711_{[.66, .76]}$	$.684_{[.61, .75]}$	$.729_{[.69, .76]}$	$.405_{[.35, .45]}$	$.334_{[.29, .37]}$
	RARe-se		$\mathbf{.705}_{[.68, .73]}$	$\mathbf{.760}_{[.70, .80]}$	$\mathbf{.685}_{[.62, .74]}$	$\mathbf{.740}_{[.70, .77]}$	$\mathbf{.495}_{[.46, .52]}$	$\mathbf{.369}_{[.32, .41]}$

5.4 Ablation Study

Table 5 addresses two questions on grid environments: whether a separate evaluation critic explains the gains of RARe, and how RARe differs structurally from Expected SARSA. Results under non-stationary dynamics are given in Appendix F.

Separate evaluation critic. With a greedy action critic, the table crosses the critic with the skip backup. Under greedy skip backups, the separate critic uses no π_e and only adds a second critic. It does not significantly change the normalized AUC in any environment relative to TempoRL, so the added critic alone does not explain the gains of RARe. The expected skip backup gives significant gains on *RCliffWalking* and *RBridge* for the shared critic and on *CliffWalking* and *RBridge* for the separate critic. With expected skip backups, RARe-se and RARe-sh reach close rewards without a consistent winner. RARe-se does achieve higher RODR on *CliffWalking*, *Bridge*, *ZigZag*, and *RBridge*, while Risk per step shows no consistent difference. Since RARe-se also uses an independent π_e schedule, this comparison reflects both the critic and the schedule.

Comparison with Expected SARSA. Expected SARSA [25] also replaces a greedy backup with an expectation, but it does so in the action critic at every step, whereas RARe keeps the action critic greedy and applies the expectation only to the skip-time bootstrap. Arms with an expected action backup therefore replace the DDQN target of the action critic with an Expected SARSA target. This replacement significantly lowers the normalized AUC in every environment, by .18 to .33 relative to RARe. With this action critic, changing the skip backup from greedy to expected has no significant effect in any environment for the shared critic. For the separate critic, it has a significant effect only in *CliffWalking* and *RBridge*. Expected skip evaluation is therefore not equivalent to Expected SARSA: the benefit depends on where the expectation is applied.

6 Related Work

Temporal abstraction. Temporal abstraction [21, 22] simplifies long-horizon tasks by enabling agents to commit to multi-step subobjectives. Semi-Markov decision processes (SMDPs) [22] are canonical formalisms, generalizing MDPs to allow decisions of variable duration. The skip paradigm specializes this idea by adaptively adjusting decision frequency, in contrast to alternative routes such as hierarchical policies [26, 13], coordinated action sequences [15, 14], and skill discovery [6, 17].

Open-loop skip. Open-loop methods optimize the temporal persistence of a single decision. DAR [11] and FiGAR [18] extend the action space with repetition lengths, and TempoRL [4] formalizes this within a Skip-MDP that decouples action and skip selection (Eq. 3). Open-loop methods reduce the number of decisions but cannot react to abrupt changes during execution. UTE [12] selects the skip length from an ensemble of skip values, using their mean plus an uncertainty term scaled by a coefficient, at the cost of additional ensemble parameters. Both TempoRL and UTE bootstrap from the greedy value at the skip's terminal state, which is the step that RARe changes.

Closed-loop skip. Closed-loop methods decide at every step whether to terminate the current action. TAAC [27] compares the value of the previous action with a newly proposed one through a softmax over the two values and learns with a multi-step compare-through operator, achieving robust performance on continuous control. This per-step gate lets it react during a repetition, but the policy must still make a decision at every step, so decision frequency is not reduced.

Risk-sensitive RL. Risk-sensitive and safe RL typically optimize a risk measure of the return or constrain costs, such as the probability of entering error states under a policy [9, 8]. This failure-based notion motivates our definition: after a skip, stochastic action selection can enter a failure state, which lowers the expected value of control at the skip's end. We therefore measure risk as this expected value loss rather than as a probability, so it needs no predefined error states, and it enters only through skip evaluation, without a risk measure or constraint in the objective.

Relation to Expected SARSA. Expected SARSA [25] replaces a sampled next action with an expectation in a one-step target, which lowers the variance of SARSA updates. RARe applies a related expectation only to the terminal state of a skip, where it evaluates a temporal commitment and can change the learned ordering over skip lengths, while keeping the action critic of the base algorithm. The two are not interchangeable: applying Expected SARSA to the action critic lowers the normalized AUC in every environment (Table 5). Preserving the optimal Skip-MDP fixed point under this modified bootstrap also requires a separate argument (Proposition 3).

7 Limitations and Future Work

RARe inherits the main limitation of open-loop repetition: it cannot react during a skip. On *HalfCheetah*, closed-loop TAAC scores higher than RARe-se (Appendix E.2), although this comparison does not isolate the loop type. Risk and RODR require known dynamics, so we measure them only on grid environments, and continuous results report reward alone. Risk is also computed from the Q^π of a frozen policy, so it measures a value gap at skip endpoints rather than realized return loss. RARe further depends on a hand-designed π_e schedule, and RARe-se is highly sensitive to it: in our schedule searches on *CliffWalking* and *ZigZag*, its performance varied widely across schedules, so the schedule needs per-environment tuning. RARe-se also adds the training cost of an evaluation critic. Finally, Proposition 3 is asymptotic, so the training-time benefits are supported empirically rather than theoretically. A natural direction is to combine open-loop and closed-loop control. An agent could commit to a skip selected by RARe and interrupt it only when continuing is no longer justified, for example when the value of the remaining skip falls below the expected value of re-deciding under π_e. This would keep the low decision frequency of open-loop repetition in benign regions while recovering the reactivity of closed-loop methods such as TAAC near hazards or under abrupt changes. The main challenge is that interrupted skips change the transitions the skip critic learns from, so skip targets must account for early termination, and the interruption check adds per-step computation. Other directions include adapting π_e to learning progress and estimating risk in continuous control.

8 Conclusion

In this work, we introduced RARe, a risk-aware action repetition framework for open-loop temporal abstraction. RARe replaces the greedy terminal bootstrap of skip-value learning with an expectation under a skip-evaluation policy π_e, keeping the base action critic and adding no inference cost. This accounts for imperfect future control during training, while preserving the optimal skip-value fixed point as π_e approaches optimality. Across six grid worlds and four continuous-control tasks, RARe performs best or comparably to the best method in reward AUC, and in grid worlds it incurs lower risk per step and makes risk-aware optimal decisions more frequently than open-loop baselines. Ablations indicate that these effects stem from the skip bootstrap rather than from Expected SARSA on the action critic or a separate critic alone. More broadly, our findings suggest that effective temporal abstraction should evaluate not only the terminal value of a commitment, but also the reliability of future control after that commitment ends.

Acknowledgments and Disclosure of Funding

This research was supported by the National Research Foundation of Korea (NRF) grants funded by the Korea government (MSIT) (Nos. RS-2022-NR068754, RS-2024-00456065 and RS-2026-25533207). Baek-Ryun Seong was supported by the AI Seoul Tech Research Support Program of the Seoul Future Foundation. The authors acknowledge the Urban Big Data and AI Institute of the University of Seoul supercomputing resources (`http://ubai.uos.ac.kr`) made available for conducting the research reported in this paper.

References

[1] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, and J. W. Vaughan, editors, *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 29304–29320, 2021.

[2] R. Bellman. A Markovian decision process. *Journal of Mathematics and Mechanics*, 6(5):679–684, 1957.

[3] C. Berner, G. Brockman, B. Chan, V. Cheung, P. Debiak, C. Dennison, D. Farhi, Q. Fischer, S. Hashme, C. Hesse, R. Józefowicz, S. Gray, C. Olsson, J. Pachocki, M. Petrov, H. P. de Oliveira Pinto, J. Raiman, T. Salimans, J. Schlatter, J. Schneider, S. Sidor, I. Sutskever, J. Tang, F. Wolski, and S. Zhang. Dota 2 with large scale deep reinforcement learning. *CoRR*, abs/1912.06680, 2019.

[4] A. Biedenkapp, R. Rajan, F. Hutter, and M. Lindauer. TempoRL: Learning when to act. In M. Meila and T. Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, Proceedings of Machine Learning Research, pages 914–924. PMLR, 2021.

[5] R. de Lazcano, K. Andreas, J. J. Tai, S. R. Lee, and J. Terry. Gymnasium Robotics. `http://github.com/Farama-Foundation/Gymnasium-Robotics`, version 1.4.0, 2024.

[6] B. Eysenbach, A. Gupta, J. Ibarz, and S. Levine. Diversity is all you need: Learning skills without a reward function. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019.

[7] C. D. Freeman, E. Frey, A. Raichuk, S. Girgin, I. Mordatch, and O. Bachem. Brax – a differentiable physics engine for large scale rigid body simulation. `http://github.com/google/brax`, version 0.14.2, 2021.

[8] P. Geibel. Reinforcement learning with bounded risk. In C. E. Brodley and A. P. Danyluk, editors, *Proceedings of the Eighteenth International Conference on Machine Learning (ICML 2001), Williams College, Williamstown, MA, USA, June 28 - July 1, 2001*, pages 162–169. Morgan Kaufmann, 2001.

[9] P. Geibel and F. Wysotzki. Risk-sensitive reinforcement learning applied to control under constraints. *J. Artif. Intell. Res.*, 24:81–108, 2005.

[10] J. Ji, B. Zhang, J. Zhou, X. Pan, W. Huang, R. Sun, Y. Geng, Y. Zhong, J. Dai, and Y. Yang. Safety Gymnasium: A unified safe reinforcement learning benchmark. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023.

[11] A. S. Lakshminarayanan, S. Sharma, and B. Ravindran. Dynamic action repetition for deep reinforcement learning. In S. Singh and S. Markovitch, editors, *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence, February 4-9, 2017, San Francisco, California, USA*, pages 2133–2139. AAAI Press, 2017.

[12] J. Lee, S. J. Park, Y. Tang, and M. Oh. Learning uncertainty-aware temporally-extended actions. In M. J. Wooldridge, J. G. Dy, and S. Natarajan, editors, *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20-27, 2024, Vancouver, Canada*, pages 13391–13399. AAAI Press, 2024.

[13] A. Levy, G. D. Konidaris, R. Platt, Jr., and K. Saenko. Learning multi-level hierarchies with hindsight. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019.

[14] Q. Li, S. Park, and S. Levine. Decoupled Q-chunking. *CoRR*, abs/2512.10926, 2025.

[15] Q. Li, Z. Zhou, and S. Levine. Reinforcement learning with action chunking. *CoRR*, abs/2507.07969, 2025.

[16] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra. Continuous control with deep reinforcement learning. In Y. Bengio and Y. LeCun, editors, *4th International Conference on Learning Representations, ICLR 2016, San Juan, Puerto Rico, May 2-4, 2016, Conference Track Proceedings*, 2016.

[17] A. Sharma, S. Gu, S. Levine, V. Kumar, and K. Hausman. Dynamics-aware unsupervised discovery of skills. In *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020*. OpenReview.net, 2020.

[18] S. Sharma, A. S. Lakshminarayanan, and B. Ravindran. Learning to repeat: Fine grained action repetition for deep reinforcement learning. In *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*. OpenReview.net, 2017.

[19] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science*, 362(6419):1140–1144, 2018.

[20] R. S. Sutton and A. G. Barto. *Reinforcement learning - an introduction*. Adaptive computation and machine learning. MIT Press, 1998.

[21] R. S. Sutton, D. Precup, and S. Singh. Intra-option learning about temporally abstract actions. In J. W. Shavlik, editor, *Proceedings of the Fifteenth International Conference on Machine Learning (ICML 1998), Madison, Wisconsin, USA, July 24-27, 1998*, pages 556–564. Morgan Kaufmann, 1998.

[22] R. S. Sutton, D. Precup, and S. Singh. Between MDPs and Semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artif. Intell.*, 112(1-2):181–211, 1999.

[23] M. Towers, A. Kwiatkowski, J. U. Balis, G. De Cola, T. Deleu, M. Goulão, A. Kallinteris, M. Krimmel, A. KG, R. Perez-Vicente, J. K. Terry, A. Pierré, S. Schulhoff, J. J. Tai, H. Tan, and O. G. Younis. Gymnasium: A standard interface for reinforcement learning environments. In D. Belgrave, C. Zhang, L. N. Montoya, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, N. Chen, I. V. M. Ruíz, and A. Loaiza-Bonilla, editors, *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*, 2025.

[24] H. van Hasselt, A. Guez, and D. Silver. Deep reinforcement learning with double Q-learning. In D. Schuurmans and M. P. Wellman, editors, *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence, February 12-17, 2016, Phoenix, Arizona, USA*, pages 2094–2100. AAAI Press, 2016.

[25] H. van Seijen, H. van Hasselt, S. Whiteson, and M. A. Wiering. A theoretical and empirical analysis of Expected Sarsa. In *IEEE Symposium on Adaptive Dynamic Programming and Reinforcement Learning, ADPRL 2009, Nashville, TN, USA, March 31 - April 1, 2009*, pages 177–184. IEEE, 2009.

[26] A. S. Vezhnevets, S. Osindero, T. Schaul, N. Heess, M. Jaderberg, D. Silver, and K. Kavukcuoglu. FeUdal networks for hierarchical reinforcement learning. In D. Precup and Y. W. Teh, editors, *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, Proceedings of Machine Learning Research, pages 3540–3549. PMLR, 2017.

[27] H. Yu, W. Xu, and H. Zhang. TAAC: temporally abstract actor-critic for continuous control. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, and J. W. Vaughan, editors, *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 29021–29033, 2021.

A Proofs for Propositions

A.1 Proof of Proposition 1

Proposition 1 (Risk in a hazard-adjacent chain)**.** *Consider a Skip-MDP containing a hazard-adjacent chain, whose hazardous states incur a negative reward when entered. Let π and π_J denote the action and skip policies, both ϵ-greedy with $\epsilon > 0$. Then, there exist a state s, an action a, and skip lengths $j_1, j_2 \in \mathcal{J}$ with $j_2 > j_1$ such that*

$$\tilde{Q}^{\pi_J}(s, j_2 \mid a) > \tilde{Q}^{\pi_J}(s, j_1 \mid a).$$

Proof. Consider a deterministic MDP with states $\{s_0, s_1, s_2, \ldots, s_j, s_{\text{haz}}\}$, where s_j is a goal terminal state with value 0, and s_{haz} is a hazardous terminal state that incurs a large negative reward $R_{\text{haz}} \ll 0$. Since this argument is independent of our proposed method, we omit the separate evaluation policy π_e and state the result directly in terms of the policy under consideration.

In each state s_i (for $i < j$), two actions are available: $\mathcal{A} = \{a, a_{\text{bad}}\}$. The transition dynamics are defined as:

- $P(s_{i+1} \mid s_i, a) = 1$
- $P(s_{\text{haz}} \mid s_i, a_{\text{bad}}) = 1$

Assume the agent follows a stochastic action policy π, such that at any s_i, it selects the bad action with probability $p \in (0, 1)$:

$$\pi(a \mid s_i) = 1 - p, \quad \pi(a_{\text{bad}} \mid s_i) = p$$

Let the reward function be $r(s, a, s_{\text{haz}}) = R_{\text{haz}}$ and 0 otherwise. Let $\gamma \in (0, 1)$ be the discount factor.

First, we observe that for any state s_i, the value $V^\pi(s_i)$ internalizes the risk of all future stochastic decisions. At each step until s_j is reached, the agent faces a probability p of hitting the hazard. The value at s_{j-1} is:

$$V^\pi(s_{j-1}) = (1-p)\gamma V^\pi(s_j) + pR_{\text{haz}} = pR_{\text{haz}} \quad (\text{since } V^\pi(s_j) = 0).$$

By backward induction, for any s_i, the value is a discounted sum of the hazard penalty, weighted by the probability of first encountering the hazard at each subsequent step. Crucially, $V^\pi(s_i) < 0$.

For the short skip ($j = 1$):

$$\begin{aligned}\tilde{Q}^{\pi_J}(s_0, 1 \mid a) &= r(s_0, a, s_1) + \gamma \mathbb{E}_{a' \sim \pi}\left[Q^\pi(s_1, a')\right] \\ &= 0 + \gamma V^\pi(s_1)\end{aligned}$$

Since $V^\pi(s_1)$ includes the cumulative risk of $j - 1$ future decision points (from s_1 to s_{j-1}), this value is significantly penalized by R_{haz}. Specifically, the probability of encountering the hazard before reaching the goal is $1 - (1-p)^{j-1}$.

For the long skip of length $j \geq 2$:

$$\begin{aligned}\tilde{Q}^{\pi_J}(s_0, j \mid a) &= \sum_{k=0}^{j-1} \gamma^k r(s_k, a, s_{k+1}) + \gamma^j \mathbb{E}_{a' \sim \pi}\left[Q^\pi(s_j, a')\right] \\ &= 0 + \gamma^j V^\pi(s_j) = 0\end{aligned}$$

Therefore:

$$\tilde{Q}^{\pi_J}(s_0, j \mid a) > \tilde{Q}^{\pi_J}(s_0, 1 \mid a)$$

which proves the claim. □

A.2 Proof of Proposition 2

Proposition 2. *Consider a deterministic MDP with a discount factor $\gamma \in (0,1)$. Let $\tilde{Q}^*$ be the optimal skip-value function, defined as:*

$$\tilde{Q}^*(s, j \mid a) = \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j V^*(s_{t+j}) \;\middle|\; s_t = s, a, j\right], \tag{5}$$

where $V^*(s) := \max_{a \in \mathcal{A}} Q^*(s,a)$. Then, for all $j \geq 1$ ($j \in \mathcal{J}$), the following inequality holds:

$$\tilde{Q}^*(s, 1 \mid a) \geq \tilde{Q}^*(s, j \mid a). \tag{6}$$

Proof. Consider a deterministic MDP with a discount factor $\gamma \in (0,1)$. For $j = 1$, we have

$$\tilde{Q}^*(s_t, 1 \mid a_t) = r_t + \gamma V^*(s_{t+1}).$$

By the Bellman optimality equation,

$$V^*(s_{t+1}) = \max_{a_{t+1}} \big(r_{t+1} + \gamma V^*(s_{t+2})\big) \;\geq\; r_{t+1} + \gamma V^*(s_{t+2}).$$

Applying this inequality recursively yields

$$V^*(s_{t+1}) \;\geq\; r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^{j-1} V^*(s_{t+j}).$$

Multiplying both sides by γ and adding r_t gives

$$r_t + \gamma V^*(s_{t+1}) \;\geq\; r_t + \gamma\left(r_{t+1} + \cdots + \gamma^{j-1} V^*\left(s_{t+j}\right)\right),$$

$$r_t + \gamma V^*(s_{t+1}) \;\geq\; r_t + \gamma r_{t+1} + \cdots + \gamma^{j} V^*(s_{t+j}),$$

which is precisely

$$\tilde{Q}^*(s_t, 1 \mid a_t) \;\geq\; \tilde{Q}^*(s_t, j \mid a_t).$$

This completes the proof. □

A.3 Proof of Proposition 3

Proposition 3. *Let π_e be the evaluation policy used in expected skip evaluation, with action-value function Q^{π_e}. Suppose $Q^{\pi_e} \to Q^*$ and that π_e converges to a deterministic optimal policy π^* such that*

$$\pi_e(a^* \mid s) \to 1 \quad \textit{for every state } s \textit{ and some } a^* \in \arg\max_{a \in \mathcal{A}} Q^*(s,a).$$

Then, for any state s, action a, and skip length $j \in \mathcal{J}$,

$$\mathbb{E}_{a' \sim \pi_e(\cdot \mid s_{t+j})}\left[Q^{\pi_e}(s_{t+j}, a')\right] \to \max_{a'} Q^*(s_{t+j}, a'). \tag{8}$$

Consequently, under TD convergence assumptions, expected skip evaluation asymptotically converges to $\tilde{Q}^*$.

Proof. For notational brevity, we present the proof assuming a finite action space $\mathcal{A}$ where the optimal action-value function Q^* is naturally bounded. Consider the expected skip evaluation target at the terminal state of a skip:

$$\mathbb{E}_{a' \sim \pi_e(\cdot \mid s_{t+j})}\left[Q^*(s_{t+j}, a')\right] = \sum_{a' \in \mathcal{A}} \pi_e(a' \mid s_{t+j}) Q^*(s_{t+j}, a').$$

Assume that π_e converges to a deterministic optimal policy π^*. Then, for some

$$a^* \in \arg\max_{a' \in \mathcal{A}} Q^*(s_{t+j}, a'),$$

we have

$$\pi_e(a^* \mid s_{t+j}) \to 1, \qquad \pi_e(a' \mid s_{t+j}) \to 0 \quad \text{for all } a' \neq a^*.$$

Since the summation features a finite number of bounded terms, the limit distributes directly into the expectation:

$$\lim_{\pi_e \to \pi^*} \sum_{a' \in \mathcal{A}} \pi_e(a' \mid s_{t+j}) Q^*(s_{t+j}, a') = Q^*(s_{t+j}, a^*).$$

Since a^* is an optimal action, by the Bellman optimality equation, we have:

$$Q^*(s_{t+j}, a^*) = \max_{a' \in \mathcal{A}} Q^*(s_{t+j}, a').$$

Thus,

$$\mathbb{E}_{a' \sim \pi_e(\cdot \mid s_{t+j})} \left[Q^*(s_{t+j}, a') \right] \to \max_{a'} Q^*(s_{t+j}, a').$$

Since $Q^{\pi_e} \to Q^*$ as $\pi_e \to \pi^*$ and $\mathcal{A}$ is finite, the same limit holds for $\mathbb{E}_{a' \sim \pi_e(\cdot \mid s_{t+j})}[Q^{\pi_e}(s_{t+j}, a')]$. Substituting this limit into the expected skip-value target gives

$$\tilde{Q}^{\pi_J}_{\text{RARe}}(s, j \mid a) \to \mathbb{E}\left[\sum_{k=0}^{j-1} \gamma^k r_{t+k} + \gamma^j \max_{a'} Q^*(s_{t+j}, a') \;\middle|\; s_t = s, a_t = a \right],$$

which is precisely the optimal skip-value target $\tilde{Q}^*(s, j \mid a)$. Hence, under standard temporal-difference convergence assumptions, expected skip evaluation asymptotically converges to $\tilde{Q}^*$. Note that for continuous action spaces, the summation is replaced by an integral, and the same result holds when $Q^*(s_{t+j}, \cdot)$ is bounded and continuous, as π_e converges weakly to a Dirac delta distribution. □

B Environment Details

B.1 Grid Environments

We evaluate RARe on six grid-world environments: CLIFFWALKING [20], BRIDGE, ZIGZAG, and their randomized versions RCLIFFWALKING, RBRIDGE, and RZIGZAG. These environments test whether an agent can select not only an appropriate action direction, but also a safe and efficient skip length under restricted optimal paths. All environments are 8×12 grids with four actions and a 100-step episode limit, with layouts illustrated in Figure 1, where green, blue, and dark red cells represent start states, goal states, and hazardous regions. The agent receives a reward of -1 for each step, including the step that reaches the goal, which ends the episode. Entering a hazardous region gives -100 and ends the episode. Observations give the agent and goal positions, and hazards are not shown. Training budgets differ across environments, from 15,000 steps on CLIFFWALKING to 200,000 on RZIGZAG (Table 7), so scores should be compared between methods within an environment rather than across environments. The maximum skip is $J = 6$ in all Grid environments.

CliffWalking provides a canonical setting in which the shortest path lies close to hazardous states. Although multiple skips may yield similar return after convergence, inappropriate decisions during training can expose the agent to severe penalties. This environment is therefore useful for evaluating whether the learned skip policy avoids unnecessary decision exposure near hazards.

Bridge requires the agent to traverse a narrow safe corridor surrounded by hazardous regions. Since deviations from the optimal path can immediately lead to failure, this environment emphasizes precise action selection and risk-aware temporal persistence. A successful skip policy must select skip lengths that advance along the bridge while avoiding over-extension into unsafe states.

ZigZag places two hazardous blocks on opposite sides of the grid, so the safe path must change direction around them. This setting tests whether the agent can adapt its skip length according to local path geometry. Long skips may be useful along straight segments, but become suboptimal near turns where replanning is required.

Randomized versions. RCLIFFWALKING, RBRIDGE, and RZIGZAG keep the base terrain and rewards, and resample the start and goal independently at each reset. RCLIFFWALKING draws them from four cells on each side of the cliff, giving 16 start–goal pairs. RBRIDGE and RZIGZAG draw them from the two outermost columns on each side, giving 256 pairs. These versions test whether a skip policy generalizes across start–goal pairs rather than memorizing one path.

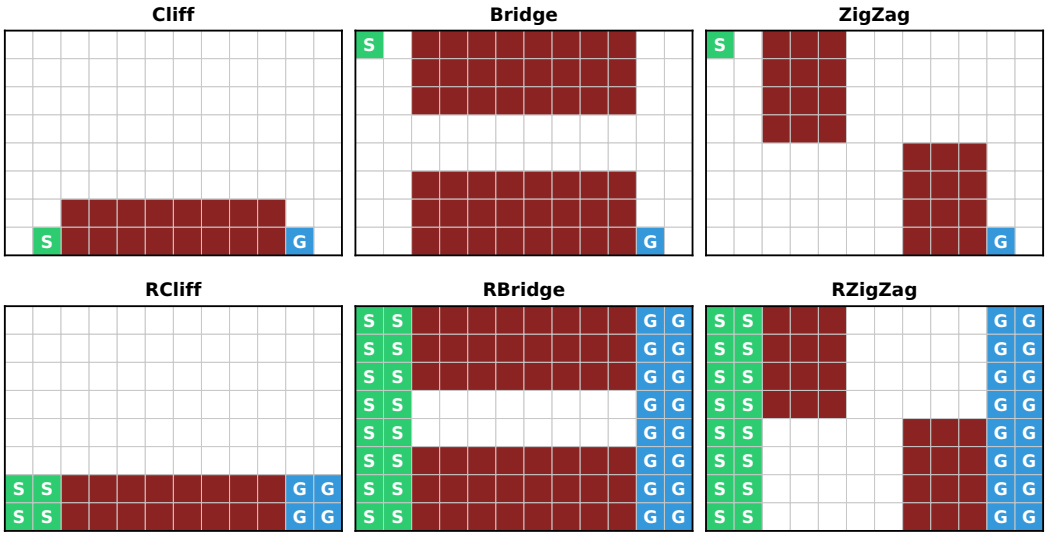


Figure 1: Grid-world environments used to evaluate risk-aware decision-making and task performance. Green, blue, and dark red cells denote start states, goals, and hazardous regions, respectively.

Table 6: Non-stationary Grid conditions. Probabilities are gust probabilities before and after the change at step 20,000.

Condition	Start and goal	Wind before	Wind after	Limit
WindShift	opposite corners, fixed	east, .10	west, .35	80
Random	Manhattan distance ≥ 9	east or north, .15	any direction, .45	90
Episodic	corner start, distance ≥ 9	east, .15	north, .35	90

B.2 Non-stationary Grid Environments

These environments test skip evaluation under changes in the dynamics. All are hazard-free 10×10 grids with four actions. After each intended move, a wind gust moves the agent one further cell in the wind direction with a given probability. The observation gives the agent position, the goal position, the wind direction, and the gust probability. The reward is -0.01 per step and 1 on reaching the goal, plus 0.1 times the decrease in Manhattan distance to the goal. All agents train for $40{,}000$ steps with $2{,}000$ warm-up steps, a replay size of $40{,}000$, a skip replay size of $120{,}000$, a maximum skip of $J = 6$, and a hidden width of 64. Evaluation runs every $1{,}000$ steps over 20 episodes, with 20 seeds for the conditions in Table 6 and 10 seeds for the shift-count sweep.

In EPISODIC, at step 12 of every episode the wind also rotates and the goal moves to the farthest other corner. The shift-count sweep extends WindShift. Initially, eastward gusts occur with probability $.10$. After the first change, gusts occur with probability $.35$, and the wind direction alternates between west and south at successive changes. Changes occur at step 20k for one shift, at 20k, 27k, and 33k for three shifts, and at 20k, 24k, 28k, 32k, and 36k for five shifts.

B.3 Continuous Environments

In continuous control, the agent chooses both an action value and how long to repeat it. The main comparison uses MountainCar and PointMaze in this section and the safety-critical Goal and Button tasks in Appendix B.4. Pendulum and HalfCheetah are used for additional analysis. Figure 2 shows all six, and Table 8 lists the settings. Training budgets again differ across environments, so scores should be compared within an environment.

MountainCar [23] requires the agent to drive an underpowered car up a steep hill by building momentum through alternating acceleration. The action is a one-dimensional force $a \in [-1, 1]$. The agent receives a reward of $+100$ on reaching the goal flag and pays a control cost of $0.1a^2$ at every

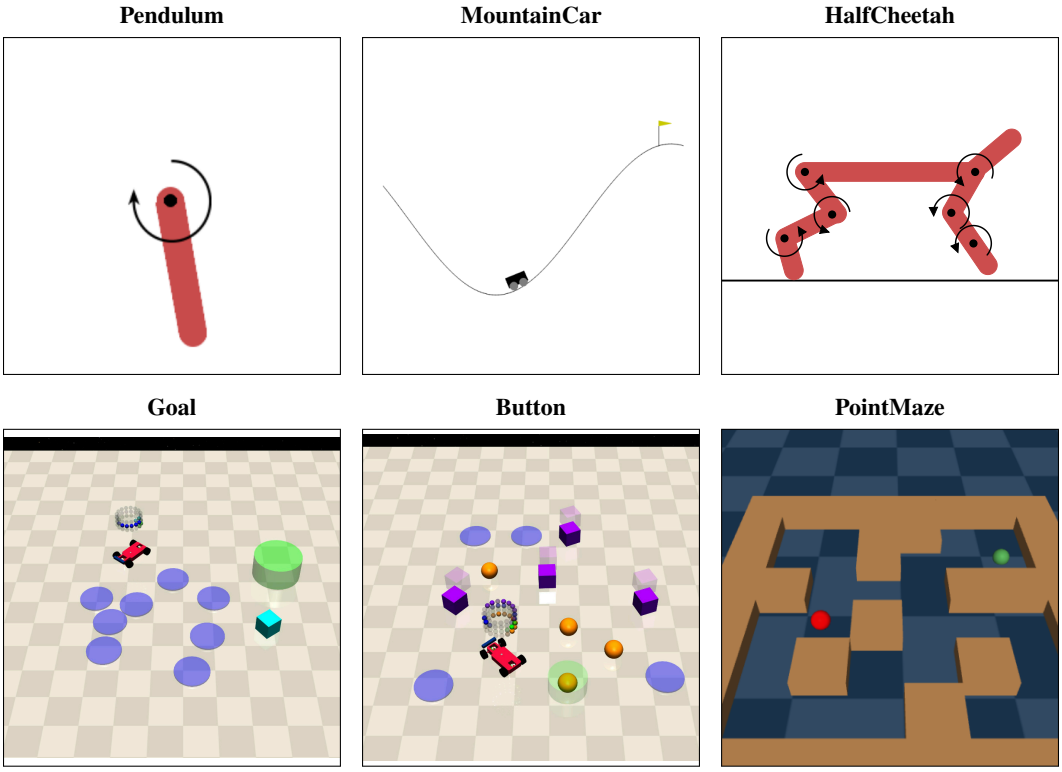


Figure 2: Overview of the continuous control environments used in our experiments. In the Pendulum, the arrow indicates the direction of applied torque around the central pivot. In the MountainCar, the black box represents the vehicle and the yellow flag marks the target location. In the HalfCheetah, the arrows mark the six actuated joints. In the Goal, the agent navigates to the goal while avoiding hazards and a movable vase. In the Button, the agent presses the target button while avoiding moving Gremlins and wrong buttons. In the PointMaze, the green sphere and the red sphere denote the agent and the goal, respectively.

step, so the goal reward is sparse. An episode ends at the goal or after 999 steps. Long skips can sustain acceleration and aid exploration, while action changes near the turning points require finer timing. This task tests whether the skip policy reduces the decision frequency without losing the timing needed to build momentum.

PointMaze [5] is a 2D navigation task in which the agent must reach a target while moving around the walls of a medium maze. The agent applies forces in the x and y directions and receives a dense reward of $\exp(-d)$ at every step, where d is the Euclidean distance to the goal. Each episode lasts 600 steps. This task evaluates long-horizon exploration: an effective skip policy should cross open areas with few decisions and make more frequent decisions near walls.

Pendulum [23] requires the agent to swing up a pendulum and balance it upright using a one-dimensional torque in $[-2, 2]$. The dense reward penalizes the angle, angular velocity, and torque at every step, and each episode lasts 200 steps. Long skips can help during the swing-up, when the torque direction changes rarely, whereas balancing near the top calls for shorter skips and finer adjustments. This task therefore tests whether the skip length adapts to the current control demand.

HalfCheetah is a planar locomotion task from Brax [7] that we simulate with its spring backend. The agent controls six joint torques, and the reward is the forward velocity minus a control cost of $0.1\|a\|^2$. Each episode lasts 1000 steps, with no early termination. Running requires coordinated control of several joints, so we use this task to examine the limits of open-loop repetition in a higher-dimensional action space.

Table 7: Grid settings. ϵ of π_e decays linearly after warm-up, over the first half of training where noted and over all of it otherwise. RARe-sh uses the behavior ϵ. The CliffWalking and ZigZag schedules were selected on the same seeds used for reporting.

Environment	Steps	Warm-up	Replay	Skip replay	Cases	ϵ of π_e (RARe-se)
CliffWalking	15,000	2,000	15,000	45,000	1	.2→.01 (first half)
Bridge	40,000	3,000	40,000	120,000	1	.2→.01
ZigZag	50,000	3,000	50,000	150,000	1	.3→.01 (first half)
RCliffWalking	50,000	2,000	50,000	150,000	16	.1→.01
RBridge	150,000	3,000	100,000	300,000	256	.2→.01
RZigZag	200,000	3,000	100,000	300,000	256	.2→.01

Table 8: Continuous settings. RARe-sh uses the behavior exploration noise as σ_e. HalfCheetah uses three seeds.

	MountainCar	PointMaze	Goal	Button	Pendulum	HalfCheetah
Training steps	50,000	200,000	52,000	65,000	17,000	3,000,000
Warm-up steps	5,000	5,000	2,000	5,000	2,000	50,000
Episode limit	999	600	1,000	1,000	200	1,000
Hidden width	64	128	64	64	64	256
Maximum skip J	10	10	5	5	10	5
Learning rate	constant	constant	constant	constant	cosine	constant
Replay size	50,000	200,000	52,000	65,000	17,000	100,000
Skip replay size	120,000	305,000	100,000	120,000	50,000	300,000
Skip ϵ decay steps	45,000	100,000	50,000	60,000	15,000	2,950,000
σ_e (RARe-se)	.1→.001	.2→0	.05→.005	.02→0	.05→.001	.05→.001
Evaluation interval	250	1,000	1,000	1,200	150	5,000
Evaluation episodes	5	4	1	1	5	3

B.4 Safety-critical Continuous Environments

The GOAL and BUTTON environments are the Level 1 Racecar tasks of the Safety-Gymnasium benchmark [10], shown in Figure 2. In both, the agent must complete its primary objective while keeping away from obstacles, and it observes nearby objects through LiDAR sensors that provide local distance and direction information. The agent is trained on the scalarized reward $w_r r - w_c c$, where r and c are the benchmark reward and cost, and $w_r = 10.0$ in both environments. Because a poorly chosen skip length can carry the agent into a hazard before it can respond, these environments are well suited to evaluating risk-aware repetition.

Goal In the GOAL environment, the agent must navigate to a target location among eight randomly placed Hazards and one Vase. It receives a dense reward for moving toward the goal and a large sparse reward upon arrival. Entering a Hazard incurs a cost, whereas the Vase is a movable object that incurs none. This task evaluates whether the agent can repeat actions in open areas while using shorter skips near Hazards. We set the cost scaling factor to $w_c = 1 \times 10^{-1}$.

Button The BUTTON task requires the agent to identify and touch a goal button among four candidates. Touching the correct button yields a goal reward and refreshes the target, and incremental rewards are given as the agent approaches it. The agent must navigate around four static Hazards and four moving Gremlins. In addition, LiDAR observations of the buttons are blocked for 10 time steps after each successful touch, so the agent must choose safe repetitions under temporary partial observability. This task evaluates efficient path generation while limiting the risk of touching incorrect buttons or colliding with moving obstacles. We set the cost scaling factor to $w_c = 5 \times 10^{-2}$.

C Algorithm Details

Shared training setup. Table 9 lists the hyperparameters, Tables 7 and 8 give the Grid and continuous budgets, and Appendix B.2 gives the non-stationary budget. Every network uses ReLU hidden activations, and every critic is trained with the Huber loss. Where used, a cosine learning rate

Table 9: Hyperparameters. Non-stationary means the non-stationary Grid environments. Dashes mark settings that do not apply.

Hyperparameter	Grid	Non-stationary	Continuous
Shared			
Optimizer	Adam	Adam	Adam
Learning rate	1×10^{-3}	1×10^{-3}	1×10^{-3}
Learning-rate schedule	cosine	constant	Table 8
Batch size	64	64	64
Discount γ	0.99	0.99	0.99
Polyak τ	0.005	0.005	0.005
Hidden layers	2	2	2
Hidden width	64	64	Table 8
Critic loss	Huber	Huber	Huber
Behavior exploration	ϵ: 1→0	ϵ: 1→.05	σ: .1→0
Open-loop methods			
Maximum skip J	6	6	Table 8
Skip exploration	behavior ϵ	behavior ϵ	Table 8
UTE			
Skip networks	10	5	10
Bootstrap mask probability	0.5	0.5	0.5
λ	-1.5	-1.5	UCB over 9 values
RARe			
π_e of RARe-sh	behavior ϵ	behavior ϵ	behavior σ
π_e of RARe-se	Table 7	ϵ: .3→.1	Table 8
Monte Carlo samples N	–	–	30
TAAC			
Hidden width	–	–	Table 8
Batch size	–	–	64
Learning rate	–	–	1×10^{-3}
Replay window	–	–	3
Initial temperature	–	–	0.1
Target entropy δ	–	–	0.05

decays to 10% of its initial value. During warm-up, actions and skip lengths are sampled uniformly. After warm-up, the behavior exploration decays linearly, over the remaining training steps on fixed Grid and continuous environments and over $18{,}000$ steps on non-stationary Grid environments. In continuous control, a random skip length is selected with a probability that decays linearly from 1 to 0.

Skip network and skip replay. All open-loop methods use the same skip network, which takes a state and an action and outputs one value per skip length, with hidden width 64 on the Grid environments and the width in Table 8 in continuous control. It has no target network, since its targets bootstrap from action-value critics. All open-loop methods also share one skip replay. A repetition is stored together with its contiguous sub-segments, each labeled as a skip of its own length.

A selected skip can exceed the steps left in the episode when the episode terminates during the repetition. The original implementations of TempoRL and UTE store only the segments that fit within the episode, so the transition of the selected length is discarded and skip lengths that extend past termination never receive an update (Figure 3, top). As a result, the skip values of these lengths remain untrained, even though the skip policy can still select them. We therefore extend the buffer: for every length from the actual end of the episode to the selected length, we store the return of the steps actually executed with a zero bootstrap discount (Figure 3, bottom). Episode time limits are treated as absorbing in the same way. We apply this extended buffer to all open-loop methods, including TempoRL and UTE, in all environments, so that the comparison isolates the skip-evaluation target rather than differences in transition storage.

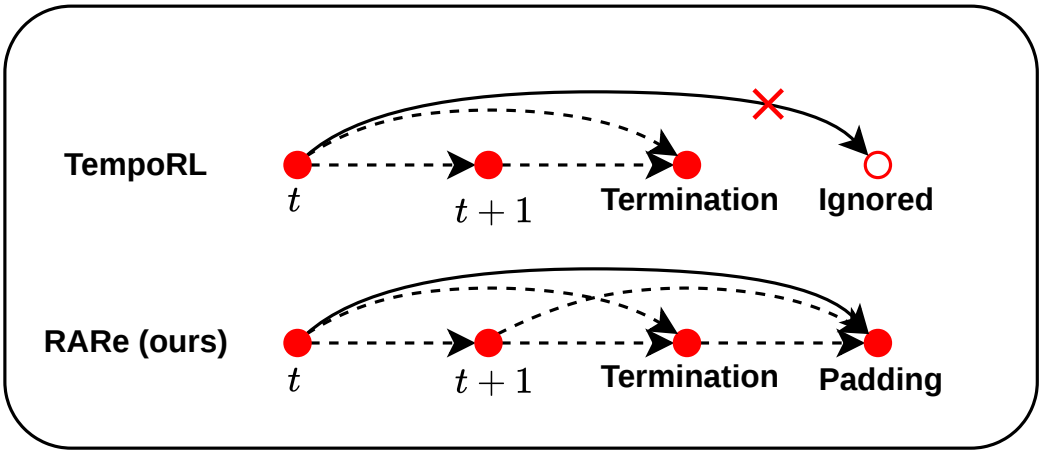


Figure 3: Original and extended skip replay. Solid arrows denote the selected skip, and dashed arrows denote its stored sub-segments. When the episode terminates during a repetition, the original buffer of TempoRL and UTE (top) stores only the segments that end at or before termination, so the transition of the selected length is discarded ($\times$) and skip lengths beyond termination receive no update. The extended buffer (bottom) pads every length from termination to the selected length with the return actually collected and a zero bootstrap discount. We apply the extended buffer to all open-loop methods, including TempoRL and UTE.

RARe. RARe-se adds an evaluation critic with the same architecture and learning rate as the base critic, used only to build training targets. RARe-sh adds no critic and uses the behavior exploration level for π_e.

DDQN [24]. We use DDQN as the fixed-frequency baseline for discrete-action environments. The agent selects a primitive action at every environment step and does not use temporal abstraction. It is also the base agent of the discrete open-loop methods.

DDPG [16]. We use DDPG as the fixed-frequency baseline for continuous-action environments. The agent outputs a continuous primitive action at every environment step and does not learn skip lengths. It is also the base agent of the continuous open-loop methods and TAAC.

TempoRL [4]. We use TempoRL as the main open-loop action repetition baseline. TempoRL learns a skip-value function and selects the skip length greedily at evaluation. Its target uses the greedy bootstrap in Eq. (3). Since TempoRL shares the repetition structure, networks, and skip replay of RARe, it provides the most direct comparison to expected skip evaluation.

UTE [12]. We include UTE as an uncertainty-aware open-loop repetition baseline. UTE trains an ensemble of skip networks with Bernoulli(0.5) bootstrap masks and selects the skip length that maximizes the ensemble mean plus λ times its standard deviation. We do not apply the multi-step Q extension. In continuous control, a UCB bandit selects λ among nine values in $[-1.5, 1.5]$, and evaluation uses its current value.

TAAC [27]. We include TAAC as a closed-loop temporal abstraction baseline in continuous control. Unlike open-loop repetition methods that commit to a selected skip length, TAAC makes a repeat-or-act decision at each step by comparing the value of continuing the previous action with the value of switching to a newly proposed action. To align it with our continuous-control setup, we convert the original SAC-based implementation to a DDPG-style actor–critic backbone, with a random stay-or-switch gate during warm-up.

Ablation. Table 10 lists the arms of Table 5. Each arm changes only the backups, and its budgets match the corresponding main runs.

Table 10: Arms of the TD-backup ablation. All arms use the budgets of Table 7. Expected SARSA uses the behavior ϵ, and expected skip backups use the π_e of the matching RARe variant.

Critic	Action backup	Skip backup	Skip bootstrap	Arm
Shared	Expected SARSA	greedy	action critic	–
Shared	Expected SARSA	expected	action critic	–
Shared	DDQN	greedy	action critic	TempoRL
Shared	DDQN	expected	action critic	RARe-sh
Separate	Expected SARSA	greedy	evaluation critic	–
Separate	Expected SARSA	expected	evaluation critic	–
Separate	DDQN	greedy	evaluation critic	–
Separate	DDQN	expected	evaluation critic	RARe-se

Table 11: Parameters on discrete environments. Training counts online and target networks, excluding optimizer states. The six Grid environments share one network size, and every ablation arm matches the RARe variant of its critic. Dashes mark methods outside the comparison.

Setting	DDQN	TempoRL	UTE	RARe-sh	RARe-se
Grid (inference)	10,628	21,642	120,768	21,642	21,642
Grid (training)	21,256	32,270	131,396	32,270	53,526
Non-stationary (inference)	–	–	31,522	10,250	10,250
Non-stationary (training)	–	–	36,454	15,182	25,046

Number of parameters. Tables 11 and 12 report parameters at inference and during training. Inference counts only the networks needed to act: the action network and skip network, plus the critic of TAAC, which compares Q-values between actions. Training also counts critics, target networks, and the RARe-se evaluation critic, and excludes optimizer states. RARe-sh and RARe-se require exactly the same number of inference parameters as TempoRL, so expected skip evaluation adds no inference overhead. During training, RARe-sh also matches TempoRL, while RARe-se adds the evaluation critic and its target network. UTE is larger because it keeps its whole ensemble, while TAAC is close to the other methods because it uses the same hidden width.

Computational cost. Tables 13 and 14 compare the cost of the methods apart from simulation. We run the actual training loop on a dummy environment that returns random observations and rewards, keeping the network and batch sizes of the RBridge and Goal runs. Jobs run one at a time on a single RTX A6000 GPU, over 5 seeds. Training time is per interaction step after warm-up and covers action selection, replay writes, and all updates, excluding compilation. Inference time is one frozen policy decision at batch size 1 inside a compiled loop; it excludes host dispatch, which takes about $90\,\mu$s for every method. RARe-sh trains at nearly the speed of TempoRL ($+0.4\%$ and $+2.3\%$), while RARe-se pays for its evaluation critic ($+17.9\%$ and $+11.5\%$) and costs about as much as UTE. Expected skip evaluation runs only during training, so a RARe decision takes the same time as a TempoRL decision and about 12% less than a UTE decision.

D Evaluation Metrics

D.1 Normalized AUC Score

To account for both learning speed and final performance, we use the normalized area under the learning curve (AUC_{norm}). In continuous control, the normalized score at training step t is

$$\text{score}_{\text{norm}}(t) = \frac{\max\left(\text{score}_{\text{agent}}(t), \text{score}_{\text{random}}\right) - \text{score}_{\text{random}}}{\text{score}_{\text{expert}} - \text{score}_{\text{random}}}, \tag{14}$$

where $\text{score}_{\text{agent}}(t)$ is the evaluation reward of the agent, and $\text{score}_{\text{random}}$ and $\text{score}_{\text{expert}}$ are the fixed references in Table 15. The expert score is a reference, not an upper bound, so the normalized score can exceed 1. On grid environments, each start–goal case c is normalized as $\text{clip}\left((R_c(t) + 100)/(R_c^* + 100), 0, 1\right)$, where $R_c(t)$ is the evaluation return and R_c^* is the optimal return of the case. On non-stationary grid environments, the score is $1 - \text{length}/\text{limit}$, where a timeout scores 0.

Table 12: Parameters on the continuous environments. Training counts online and target networks, excluding optimizer states. HalfCheetah compares TAAC and RARe-se only.

Model	MountainCar	PointMaze	Goal	Button	Pendulum	HalfCheetah
Inference						
DDPG	4,417	17,922	8,194	9,218	–	–
TAAC	8,962	36,227	16,579	18,627	9,090	145,671
TempoRL	9,483	37,132	16,711	18,759	–	–
UTE	55,077	210,022	93,364	104,628	–	–
RARe-sh	9,483	37,132	16,711	18,759	–	–
RARe-se	9,483	37,132	16,711	18,759	9,611	145,163
Training						
DDPG	17,796	71,942	32,902	36,998	–	–
TAAC	17,925	72,455	33,159	37,255	18,181	291,343
TempoRL	22,862	91,152	41,419	46,539	–	–
UTE	68,456	264,042	118,072	132,408	–	–
RARe-sh	22,862	91,152	41,419	46,539	–	–
RARe-se	31,824	127,250	57,933	65,101	32,272	505,877

Table 13: Discrete method cost on dummy data with the RBridge network sizes, mean $\pm$ SD over 5 seeds. Simulation is excluded.

Model	Training (ms/step)	vs. TempoRL	vs. UTE	Inference (μs/decision)
DDQN	0.178 ± 0.004	-53.1%	-59.7%	8.12 ± 0.01
TempoRL	0.379 ± 0.002	–	-14.1%	12.82 ± 0.02
UTE	0.441 ± 0.002	$+16.4\%$	–	14.64 ± 0.27
RARe-sh	0.381 ± 0.002	$+0.4\%$	-13.7%	12.82 ± 0.02
RARe-se	0.447 ± 0.001	$+17.9\%$	$+1.3\%$	12.81 ± 0.02

Evaluation runs without exploration, every 1% of training on grid environments and on every start–goal pair of the randomized versions; other schedules are given in Table 8 and Appendix B.2. For each seed, AUC_{norm} integrates the score over evaluation checkpoints $t_1 < \cdots < t_K$ with the trapezoidal rule:

$$\text{AUC}_{\text{norm}} = \frac{1}{t_K - t_1} \sum_{k=1}^{K-1} (t_{k+1} - t_k) \, \frac{\text{score}_{\text{norm}}(t_k) + \text{score}_{\text{norm}}(t_{k+1})}{2}. \tag{15}$$

Checkpoints start after warm-up, except on non-stationary grid environments, where they cover training steps 20k–40k, which start at the first change. Randomized grid environments average the AUCs of their cases. Risk and RODR are summarized with the same integral. Paired comparisons resample seeds jointly with $10{,}000$ bootstrap resamples and report the difference of interquartile means with its 95% confidence interval.

D.2 Risk per Step

Risk per step is computed on grid environments. At each checkpoint, we freeze the behavior ϵ-greedy policy π of the base critic and compute its exact finite-horizon action values Q^π from the known dynamics, so no learned value enters the metric. Let $\mathcal{T}$ be the evaluation episodes at the checkpoint, each with decisions $\{(s_n, a_n, j_n)\}_{n=1}^{N_\tau}$ under the deterministic evaluation policy and T_τ primitive steps. Risk per step is

$$\text{Risk}_{\text{step}} = \frac{\sum_{\tau \in \mathcal{T}} \sum_{n=1}^{N_\tau} \textit{Risk}(s_n, a_n, j_n)}{\sum_{\tau \in \mathcal{T}} T_\tau}, \tag{16}$$

where *Risk* uses $Q = Q^\pi$ in Eq. (4) and is zero when the skip ends at a goal, a hazard, or the time limit. The same construction is used for all skip methods. DDQN does not repeat actions, so this metric is not reported for it. We divide by primitive steps because a per-episode sum would fall whenever an episode ends early in a hazard.

D.3 Risk-aware Optimal Decision Ratio

Table 14: Continuous method cost on dummy data with the Goal network sizes, mean ± SD over 5 seeds. Simulation is excluded.

Model	Training (ms/step)	vs. TempoRL	vs. UTE	Inference (μs/decision)
DDPG	0.228 ± 0.003	-67.5%	-71.0%	7.37 ± 0.02
TAAC	0.319 ± 0.002	-54.3%	-59.3%	7.52 ± 0.02
TempoRL	0.700 ± 0.012	–	-10.9%	12.20 ± 0.04
UTE	0.785 ± 0.014	$+12.3\%$	–	13.85 ± 0.05
RARe-sh	0.715 ± 0.014	$+2.3\%$	-8.9%	12.19 ± 0.02
RARe-se	0.780 ± 0.011	$+11.5\%$	-0.7%	12.16 ± 0.04

Table 15: Reference scores of the normalized AUC. On grid environments, the expert score is the optimal return of each start–goal case, so randomized versions show the range over cases. The HalfCheetah expert is the final score of a Brax PPO reference and is not an upper bound.

	CliffWalking	Bridge	ZigZag	RCliffWalking	RBridge	RZigZag
Floor	-100	-100	-100	-100	-100	-100
Expert	-13	-17	-19	$[-15, -11]$	$[-18, -9]$	$[-20, -10]$
	MountainCar	PointMaze	Goal	Button	Pendulum	HalfCheetah
Random	-1.7	17	-6	-12	-1400	-65.5
Expert	99	400	100	50	-173	4936

The Risk-aware Optimal Decision Ratio (RODR) evaluates whether the learned policies make decisions consistent with risk-aware optimal behavior. Unlike return-based metrics, it inspects each decision and checks the action and the skip length jointly.

Let $d(s)$ be the shortest safe distance from s to the goal. The optimal action set $\mathcal{A}^*(s)$ contains the actions that move to a non-hazardous state with smaller d. For each state s and action $a^* \in \mathcal{A}^*(s)$, the oracle dictionary $\mathcal{D}^*(s, a^*)$ contains the skip lengths that make maximal safe progress. We repeat a^* up to $J = 6$ steps, stopping before any hazard, and keep the length whose endpoint has the smallest d. If the repetition reaches the goal, all lengths that reach it are valid. All methods are scored with the same oracle.

A decision can match the oracle while only recovering ground it has already lost, so we also require new progress. Given an evaluation trajectory $\tau = \{(s_n, a_n, j_n)\}_{n=1}^{N_\tau}$, let e_n be the endpoint of decision n and $\mathcal{V}_n$ the set of states visited before e_n in the episode. RODR is

$$\mathrm{RODR}(\tau) = \frac{1}{N_\tau} \sum_{n=1}^{N_\tau} \mathbb{I}\left(a_n \in \mathcal{A}^*(s_n) \ \wedge \ j_n \in \mathcal{D}^*(s_n, a_n) \ \wedge \ d(e_n) < \min_{s \in \mathcal{V}_n} d(s) \right).$$

The value at a checkpoint is the mean over its evaluation episodes. The procedure is summarized in Algorithm 2.

Relation to Risk. The two metrics answer different questions. Risk is value-based: it measures how much value the stochastic policy would lose where control returns, but not whether the agent is making progress. RODR is geometric: it does not depend on Q or ϵ, and it measures whether each decision is correct, but not how costly the endpoint is. An agent can lower Risk by stopping at safe but unproductive states, and a decision credited by RODR can still end at a state with high Risk. As discussed in Proposition 1, shorter skips expose the agent to additional decision points. Reporting both metrics shows whether a method reduces this exposure while preserving progress along the optimal path.

E Learning Curves

We report the curves behind the AUC tables in the main text. In every figure, the line is the IQM over 20 seeds at each post-warm-up checkpoint, and the shaded band is its 95% bootstrap confidence interval. Curves are smoothed with a moving average of five checkpoints for readability, whereas the tables use the unsmoothed values.

Algorithm 2 Risk-aware Optimal Decision Ratio

Require: Evaluation trajectories $\tau_1, \ldots, \tau_M$ of one seed at one checkpoint, optimal action sets $\mathcal{A}^*$, oracle dictionary $\mathcal{D}^*$, shortest safe distance d
Ensure: RODR at the checkpoint

Initialize $\rho \leftarrow 0$
for $m = 1$ to M **do**
  Let $\tau_m = \{(s_n, a_n, j_n)\}_{n=1}^{N_m}$
  $C \leftarrow 0$, $\mathcal{V} \leftarrow \{s_1\}$
  for $n = 1$ to N_m **do**
    Let e_n be the endpoint of decision n, and add the states passed before e_n to $\mathcal{V}$
    if $a_n \in \mathcal{A}^*(s_n)$ **and** $j_n \in \mathcal{D}^*(s_n, a_n)$ **and** $d(e_n) < \min_{s \in \mathcal{V}} d(s)$ **then**
      $C \leftarrow C + 1$
    end if
    $\mathcal{V} \leftarrow \mathcal{V} \cup \{e_n\}$
  end for
  $\rho \leftarrow \rho + C/N_m$
end for
return ρ/M

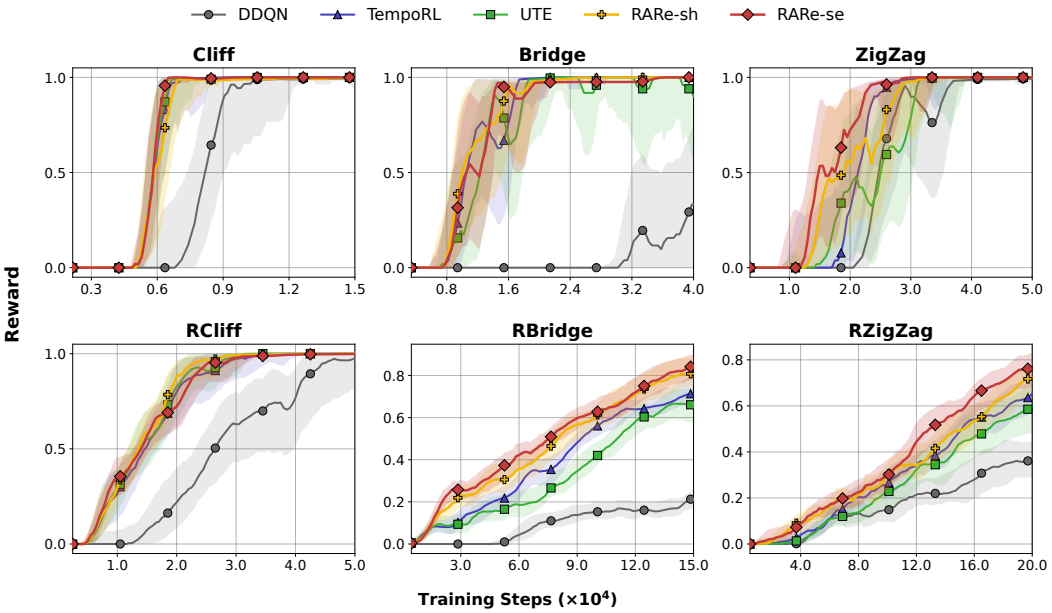


Figure 4: Normalized reward on grid environments during training.

E.1 Grid Environments

Figures 4, 5, and 6 show the normalized reward, Risk per step, and RODR during training. Risk per step falls late in training for all methods because it is computed under the behavior policy, whose ϵ decays to zero. The gap in Eq. (4) then vanishes except on *RBridge* and *RZigZag*, where the learned greedy action remains suboptimal under Q^π on some states. The comparison between methods is therefore most informative in the middle of training, when future control is still imperfect.

E.2 Continuous Environments

Figure 7 shows the normalized reward during training. Scores can exceed 1 because the expert score is a reference, not an upper bound. Table 16 compares closed-loop TAAC with RARe-se: RARe-se scores higher on *Pendulum*, whereas TAAC scores higher on *HalfCheetah*. The two tasks differ in both action dimension and budget, so this comparison does not isolate the effect of the loop type.

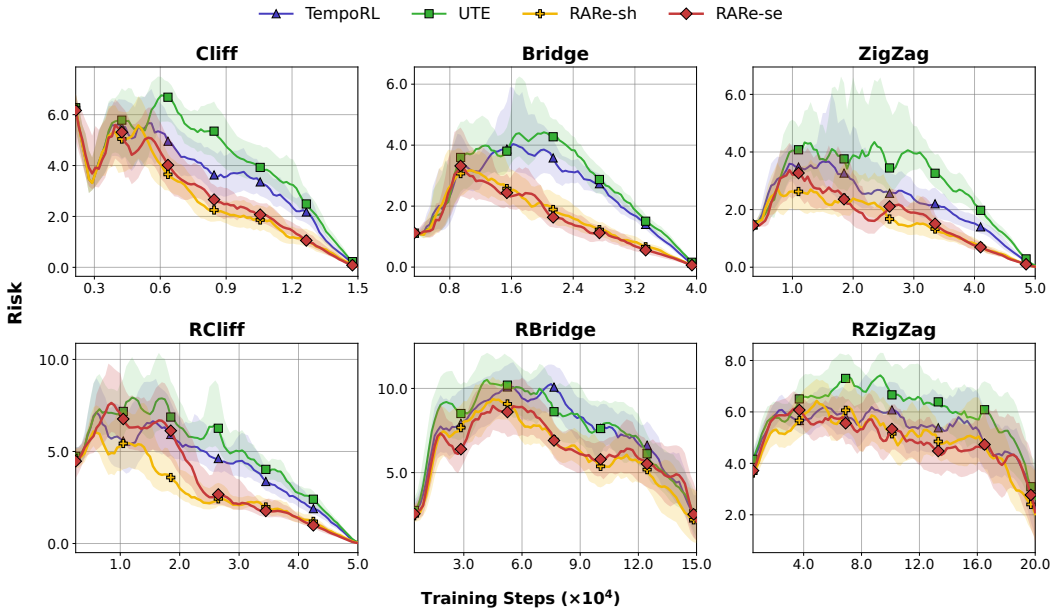


Figure 5: Risk per step on grid environments during training (lower is better). DDQN does not repeat actions, so it is not shown.

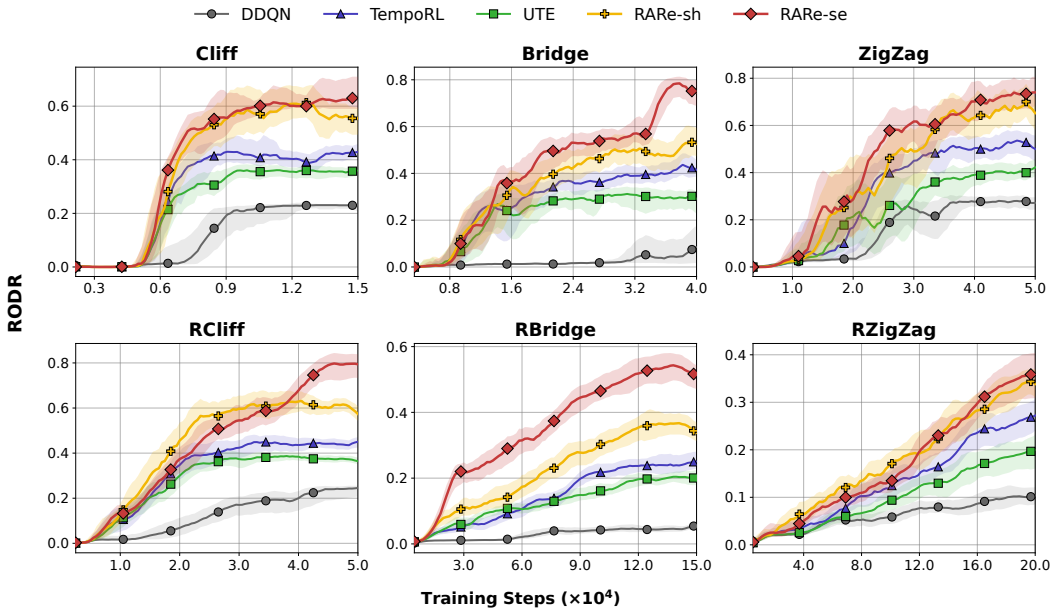


Figure 6: RODR on grid environments during training.

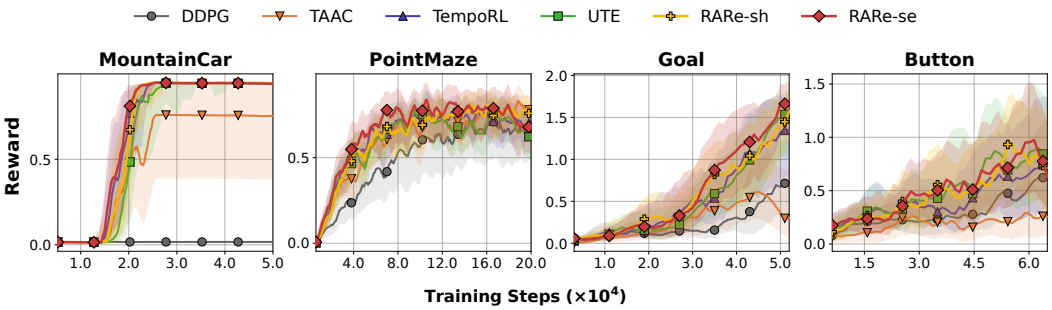


Figure 7: Normalized reward on continuous environments during training.

Table 16: Normalized reward AUC of closed-loop TAAC and open-loop *RARe* across action dimensions (higher is better). Pendulum: IQM over 20 seeds with 95% bootstrap CI. HalfCheetah: mean over 3 seeds with the seed range. Reference scores are listed in Table 15.

Loop	Model	Pendulum	HalfCheetah
Action dim.		1	6
Steps		17k	3M
Closed	TAAC	$.848_{[.83, .86]}$	$\mathbf{.605}_{[.53, .68]}$
Open	RARe-se	$\mathbf{.877}_{[.86, .89]}$	$.250_{[.21, .27]}$

F Non-stationary Results

We test whether expected skip evaluation remains useful when the dynamics change during training. We compare RARe with UTE, whose uncertainty-based skip selection is a natural alternative in this setting, on the hazard-free windy grids of Appendix B.2. As shown in Table 17, both RARe variants match or exceed UTE in all six conditions. The paired 95% confidence interval excludes zero for RARe-se on *Random* and three shifts, and for RARe-sh at one, three, and five shifts. Differences on *WindShift* and *Episodic* are not significant, and neither RARe variant significantly outperforms the other.

Table 17: Post-change AUC of $1 - \text{length}/\text{max steps}$ over 20k–40k steps on non-stationary grids (IQM with 95% bootstrap CI; higher is better). Environment columns use 20 seeds and the shift-count sweep uses 10.

Model	Environment			Number of Shifts		
	WindShift	Random	Episodic	1	3	5
UTE	$.593_{[.56, .63]}$	$.516_{[.49, .54]}$	$.400_{[.38, .42]}$	$.595_{[.54, .64]}$	$.511_{[.43, .56]}$	$.545_{[.46, .60]}$
RARe-sh	$.605_{[.58, .63]}$	$.539_{[.51, .57]}$	$.402_{[.38, .42]}$	$\mathbf{.653}_{[.60, .68]}$	$\mathbf{.580}_{[.52, .63]}$	$\mathbf{.599}_{[.56, .63]}$
RARe-se	$\mathbf{.631}_{[.60, .65]}$	$\mathbf{.558}_{[.53, .59]}$	$\mathbf{.406}_{[.38, .44]}$	$.610_{[.56, .65]}$	$.569_{[.51, .63]}$	$.581_{[.53, .62]}$

G Compute Resources

All experiments were conducted on a server equipped with eight NVIDIA RTX A6000 GPUs with 48GB of memory each, of which at most four were used at a time. Each run used a single GPU. The system used NVIDIA driver version 545.23.08 with CUDA 12.3. The CPU was a dual-socket AMD EPYC 7763 machine with 128 physical CPU cores in total, running at 2.45GHz. The machine contains two NUMA nodes, each with 64 CPU cores, and 1TB of memory.

The main hardware specifications are summarized as follows:

- **GPU**: NVIDIA RTX A6000, 48GB memory.
- **CUDA**: CUDA 12.3 with NVIDIA driver 545.23.08.

- **CPU**: AMD EPYC 7763 CPU @ 2.45GHz.
- **CPU cores**: 128 physical cores across two sockets.
- **NUMA configuration**: 2 NUMA nodes, each with 64 CPU cores.
- **Memory**: 1TB.
- **Software**: Python 3.11, JAX 0.4.38, Ubuntu 22.04.

All reported experiments were run using this compute setup unless otherwise specified. The wall-clock time varies across environments and baselines: grid-world runs finish within an hour, while continuous-control and Safe RL runs take up to a few hours due to simulator cost. The 3M-step HalfCheetah runs take about ten hours. Each main result is computed over 20 random seeds per method; the shift-count sweep uses 10 seeds and HalfCheetah uses 3. Seeds are batched in one process, except that Goal and Button run in four shards of five seeds. RARe's expected-target computation is used only during training and does not add inference-time computation.

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